PROJECT: HYPOTHESIS TESTING WITH MEN'S AND WOMEN'S SOCCER MATCHES

datalab

You're working as a sports journalist at a major online sports media company, specializing in soccer analysis and reporting. You've been watching both men's and women's international soccer matches for a number of years, and your gut instinct tells you that more goals are scored in women's international football matches than men's. This would make an interesting investigative article that your subscribers are bound to love, but you'll need to perform a valid statistical hypothesis test to be sure!

While scoping this project, you acknowledge that the sport has changed a lot over the years, and performances likely vary a lot depending on the tournament, so you decide to limit the data used in the analysis to only official `FIFA World Cup` matches (not including qualifiers) since `2002-01-01`.

You create two datasets containing the results of every official men's and women's international football match since the 19th century, which you scraped from a reliable online source. This data is stored in two CSV files: `women_results.csv` and `men_results.csv`.

The question you are trying to determine the answer to is:

> Are more goals scored in women's international soccer matches than men's?

You assume a **10% significance level**, and use the following null and alternative hypotheses:

H_0 : The mean number of goals scored in women's international soccer matches is the same as men's.

H_A : The mean number of goals scored in women's international soccer matches is greater than men's.

```
# Start your code here!
import pandas as pd
import matplotlib.pyplot as plt
import pingouin
from scipy.stats import mannwhitneyu
```

```
mens = pd.read_csv("men_results.csv")
womens = pd.read_csv("women_results.csv")
```

```
mens.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 44353 entries, 0 to 44352
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  44353 non-null  int64
 1   date        44353 non-null  object
 2   home_team   44353 non-null  object
 3   away_team   44353 non-null  object
 4   home_score  44353 non-null  int64
 5   away_score  44353 non-null  int64
 6   tournament  44353 non-null  object
dtypes: int64(3), object(4)
memory usage: 2.4+ MB
```

```
womens.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4884 entries, 0 to 4883
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  4884 non-null   int64
 1   date        4884 non-null   object
 2   home_team   4884 non-null   object
 3   away_team   4884 non-null   object
 4   home_score  4884 non-null   int64
 5   away_score  4884 non-null   int64
 6   tournament  4884 non-null   object
dtypes: int64(3), object(4)
memory usage: 267.2+ KB
```

```
mens.sample(5)
```

...	Unn...	date	home_team	away_team	ho...	aw...	tournament
33377	33377	2011-05-24	Republic of Ireland	Northern Ireland	5	0	Nations Cup
15805	15805	1989-05-20	German DR	Austria	1	1	FIFA World Cup q
3505	3505	1953-06-08	Guinea-Bissau	Gambia	1	0	Friendly
23999	23999	2001-02-28	Liechtenstein	Latvia	0	2	Friendly
22439	22439	1999-06-24	Paraguay	Colombia	2	1	Friendly

Rows: 5 Expand

```
mens['date'] = pd.to_datetime(mens['date'])
womens['date'] = pd.to_datetime(womens['date'])
```

```
mens['tournament'].value_counts()
```

index	tournament
Friendly	
FIFA World Cup qualification	
UEFA Euro qualification	
African Cup of Nations qualification	
FIFA World Cup	
Copa América	
AFC Asian Cup qualification	
African Cup of Nations	
CECAFA Cup	
CFU Caribbean Cup qualification	
Merdeka Tournament	
British Home Championship	
UEFA Nations League	
Gulf Cup	
AFC Asian Cup	
Gold Cup	

Rows: 141 Expand

```
womens['tournament'].value_counts()
```

index	t...
UEFA Euro qualification	1445
Algarve Cup	551
FIFA World Cup	284
AFC Championship	268
Cyprus Cup	258
African Championship qualification	226
UEFA Euro	184
African Championship	173
FIFA World Cup qualification	172
CONCACAF Gold Cup qualification	143
AFC Asian Cup qualification	141
Copa América	131
Olympic Games	130
CONCACAF Gold Cup	126
Friendly	111
AFC Asian Cup	111

Rows: 30 Expand

```
category = ['FIFA World Cup']
```

```
mens_filtered = mens[(mens['date'] > '2002-1-1') & (mens['tournament'].isin(category))]
```

```
mens_filtered.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 384 entries, 25164 to 44352
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  384 non-null    int64
 1   date        384 non-null    datetime64[ns]
 2   home_team   384 non-null    object
 3   away_team   384 non-null    object
 4   home_score  384 non-null    int64
 5   away_score  384 non-null    int64
 6   tournament  384 non-null    object
dtypes: datetime64[ns](1), int64(3), object(3)
memory usage: 24.0+ KB
```

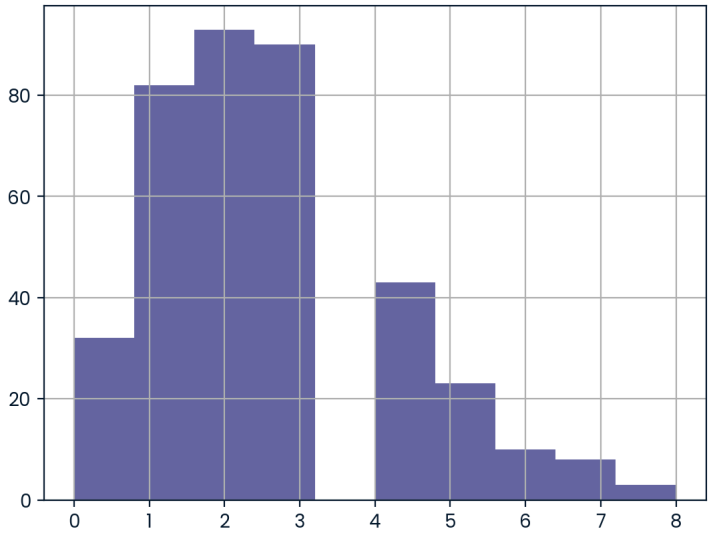
```
womens_filtered = womens[(womens['date'] > '2002-1-1') & (womens['tournament'].isin(category))]
```

```
womens_filtered.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 200 entries, 1600 to 4469
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Unnamed: 0  200 non-null    int64
 1   date        200 non-null    datetime64[ns]
 2   home_team   200 non-null    object
 3   away_team   200 non-null    object
 4   home_score  200 non-null    int64
 5   away_score  200 non-null    int64
 6   tournament  200 non-null    object
dtypes: datetime64[ns](1), int64(3), object(3)
memory usage: 12.5+ KB
```

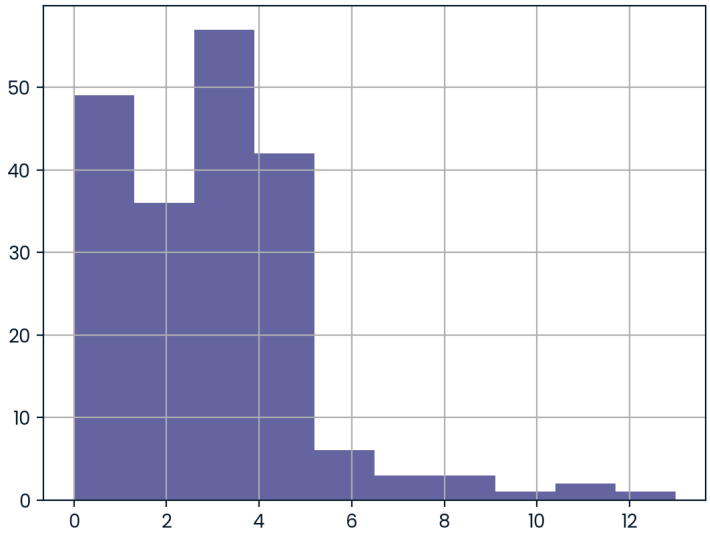
```
mens_filtered['group'] = 'men'
womens_filtered['group'] = 'women'
mens_filtered['goals_scored'] = mens_filtered['home_score'] + mens_filtered['away_score']
womens_filtered['goals_scored'] = womens_filtered['home_score'] + womens_filtered['away_score']
```

```
mens_filtered["goals_scored"].hist()
plt.show()
plt.clf()
```




```
<Figure size 640x480 with 0 Axes>
```

```
womens_filtered["goals_scored"].hist()
plt.show()
plt.clf()
```




```
<Figure size 640x480 with 0 Axes>
```

```
u_stat, p_value = mannwhitneyu(
    mens_filtered['goals_scored'],
    womens_filtered['goals_scored'],
    alternative='two-sided'
)

print("U statistic:", u_stat)
print("p-value:", p_value)
```

```
U statistic: 33527.0
p-value: 0.010213219650887282
```

```
mens_filtered.head()
```

...	U...	date	home_team	awa...	h...	a...	tournam...
25164	25164	2002-05-31T00:00:00.000	France	Senegal	0	1	FIFA World Cup
25165	25165	2002-06-01T00:00:00.000	Germany	Saudi Arabia	8	0	FIFA World Cup
25166	25166	2002-06-01T00:00:00.000	Republic of Ireland	Cameroon	1	1	FIFA World Cup
25167	25167	2002-06-01T00:00:00.000	Uruguay	Denmark	1	2	FIFA World Cup
25168	25168	2002-06-02T00:00:00.000	Argentina	Nigeria	1	0	FIFA World Cup

Rows: 5 Expand

```
both = pd.concat([womens_filtered, mens_filtered], axis=0, ignore_index=True)
```

```
both.sample(10)
```

...	U...	date	h	away_team	h...	a...	tournam...	
521	44284	2022-11-21T00:00:00.000	Senegal	Netherlands	0	2	FIFA World Cup	me
419	36530	2014-06-21T00:00:00.000	Germany	Ghana	2	2	FIFA World Cup	me
520	44283	2022-11-20T00:00:00.000	Qatar	Ecuador	0	2	FIFA World Cup	me
56	2162	2007-09-22T00:00:00.000	Germany	North Korea	3	0	FIFA World Cup	wo
26	1631	2003-10-02T00:00:00.000	Germany	Russia	7	1	FIFA World Cup	wo
555	44318	2022-11-29T00:00:00.000	Iran	United States	0	1	FIFA World Cup	me
214	25178	2002-06-05T00:00:00.000	Germany	Republic of Ireland	1	1	FIFA World Cup	me
297	28771	2006-06-20T00:00:00.000	Germany	Ecuador	3	0	FIFA World Cup	me
387	32606	2010-07-03T00:00:00.000	Paraguay	Spain	0	1	FIFA World Cup	me
105	3435	2015-06-09T00:00:00.000	Brazil	South Korea	2	0	FIFA World Cup	wo

Rows: 10 Expand

```
both_subset = both[["goals_scored", "group"]]
both_subset_wide = both_subset.pivot(columns="group", values="goals_scored")
```

```
both_subset_wide.head(10)
```

...	...	...
0		3
1		2
2		5
3		6
4		4
5		3
6		3
7		1
8		5
9		1

Rows: 10 Expand

```
results_pg = pingouin.mwu(x=both_subset_wide["women"],
                          y=both_subset_wide["men"],
                          alternative="greater")
```

```
print(results_pg)
```

```
       U-val alternative     p-val       RBC      CLES
MWU  43273.0     greater  0.005107 -0.126901  0.563451
```

```
results_scipy = mannwhitneyu(x=womens_filtered["goals_scored"],
                             y=mens_filtered["goals_scored"],
                             alternative="greater")
```

```
print(results_scipy)
```

```
MannwhitneyuResult(statistic=43273.0, pvalue=0.005106609825443641)
```

```
p_val = results_pg["p-val"].values[0]
```

```
if p_val <= 0.1:
    result = "reject"
else:
    result = "fail to reject"

result_dict = {"p_val": p_val, "result": result}
```

```
result_dict
```

```
{'p_val': 0.005106609825443641, 'result': 'reject'}
```

Write Python code or tell our AI what to do